Geotagged Airline Tweets

Elasticsearch indexing and query activity

A. Dataset exploration

The Twitter US Airline Sentiment dataset contains 14,640 tweet rows. Dataset exploration reports 1,019 non-empty *tweet_coord* values, including 164 placeholder coordinates equal to [0.0, 0.0]. Those placeholder values are excluded from the geographic index; 855 non-empty rows remain as valid coordinate points.

Source: Hugging Face dataset card and an initial-observations analysis of the original Tweets.csv file. The attached starter archive contains only a five-row sample, not the full dataset.

B. Indexer and bulk load

The completed Python indexer creates the tweets index with strict mappings, reads the CSV one row at a time, cleans empty fields, unescapes HTML entities, extracts lower-case hashtags and mentions, converts timestamps to ISO 8601, and maps valid coordinates to geo_point values. It uses helpers.streaming_bulk with chunks of 500 and tweet_id as the document ID.

Check	Result
Sample dry-run	Expected: 5 rows read, 0 skipped, 2 valid coordinates. Row 5 [0.0, 0.0] is omitted as a location.
Full CSV index run	Rows read: _____ Indexed: _____ Failed: _____ Documents in tweets: _____
GET /tweets/_count	Record the count returned by Kibana after indexing: _____

Why Bulk API, tweet_id, and coordinate cleanup? Bulk indexing sends many records in fewer requests, which reduces request overhead and makes large imports faster. Using tweet_id as Elasticsearch _id makes reruns idempotent because a tweet updates the same document instead of creating a duplicate. The [0.0, 0.0] points are placeholders rather than real GPS positions, so removing them prevents false geographic matches and misleading map results.

C. Query results worksheet

Run the corresponding entries in queries.txt in Kibana Dev Tools after indexing the full CSV. Record the counts, buckets, or sample tweets returned by your Elasticsearch instance. These values depend on the live index and are intentionally left blank here.

Query	Result to record
Q1 Negative total; United negative total	_____
Q2 Five matching luggage/bag tweets	_____
Q3 Positive exact-phrase count	_____
Q4 Matching Delta tweets	_____
Q5 Five highest retweet counts	_____
Q6 Airline totals and sentiment buckets	_____
Q7 Top negative reasons by airline	_____
Q8 Daily counts and busiest date	_____
Q9 Ten frequent hashtags	_____
Q10 Geotag count and share	_____
Q11 JFK matches and nearest distances	_____
Q12 California negative counts by airline	_____
Q13 Chicago distance-ring counts	_____
Q14 Five busiest map cells and negative centroid	_____

Submission files: tweets_indexer.py, queries.txt, and this report.pdf. Keep Tweets.csv in the same working directory when running the indexer. The full dataset was not included in the provided starter archive, and no Elasticsearch cluster was connected when this report was prepared.